RECAP

"QUICK!"

(initially it was a "5 minute adventure")

NETWORKED CONTROL SYSTEM

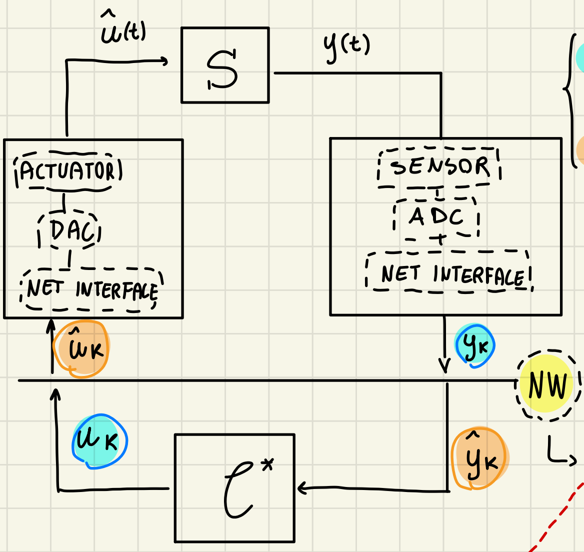
↳ probleems arising in systems where ⇒ lot of actuators / sensors comunicate through a shared comunication channel (network bus)

this cause communication issues:

- BANDWIDTH LIMITATIONS
- PACKET DELAYS
- PACKET DROPOUTS

We focus on the effect of this drawback of NCS on the stability on the controlled system, (assume to have a Network design given + communication protocol)

↳ CONTROL UNDER NON-IDEAL COMMUNICATION



DATA RECORDS = the ones transmitted to the network

DATA IMAGE = the ones that NW transmitt to destination

⇓

We focus on the difference (induced delay / possible dropouts) between records and image, and the effect of it on the NCS

↳ even if we have the listed issues, a shared bus NW comes with lots of advantages when more nodes comunicate on same net, with respect to a point-to-point communication

- Modularity
  - distribute operations over nodes in NET
  - interoperability
  - Reconfigurability
  - simple plug & play
  - self configuration controller
- Diagnostic
- Avoid single point-of-failure

- BANDWIDTH LIMITATIONS limit in the amount of bit / packet that could be sent, causing the classical segmentation trade-off issue

  (small Tsample, but more packets / big Tsample, but less packets)

- PACKET DELAYS due to stochastic sources of delays, related to waiting time, net time, destination time... can affect the NCS expecially in critical real time application

- PACKET DROPOUTS due to failure in the nodes OR in the medium OR because of too large waiting time on the buffers → NO more real-time

the aim of CONTROL NETWORKS, is to support Network control system with a proper transmission of data, usually small data (sensors/ actuator data) but with higher frequency (more data to be sent)
With guarantee of
- Reliable communication
- bounded (constant as possible) delay

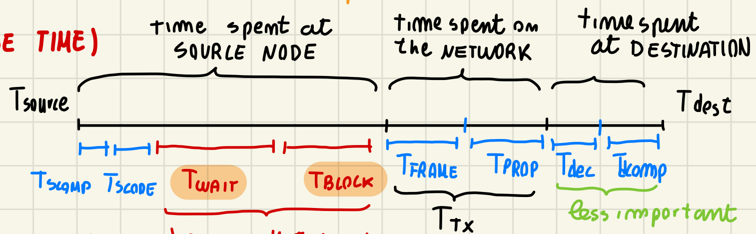
} much more in safety critical applications

- In particular the problems in communication arises when the Network is congested

↳ congestion is related to the used MAC protocol (MEDIA ACCESS CONTROL)

TIME DELAY (RESPONSE TIME)

$T_{delay} = T_{dest} - T_{source}$



(MOST relevant, it depends on MAC) ← here packet wait on the buffer to be sent, first wait for its turn, then when ready to be sent it waits due to possible network occupancy

T_{delay}

$T_{tx} = T_{PROP} + T_{FRAME}$

T_{PROP}: time to propagate on the phisical medium

T_{FRAME}: time to sent all the packet bits on the Network, relate to the BITRATE R [bit/s] and to PACKET LENGHT N_B [bit]

$$T_{PROP} = \frac{\text{channel lenght}}{v_{PROP}}$$

$$T_{FRAME} = \frac{N_B}{R}$$

where $R_{MAX} = B \log_2(1 + S/N)$ for shannon's theorem

B: channel bandwidth

S/N: signal to noise ratio in linear scale

⇘ but usually $R \ll R_{MAX}$

⇓

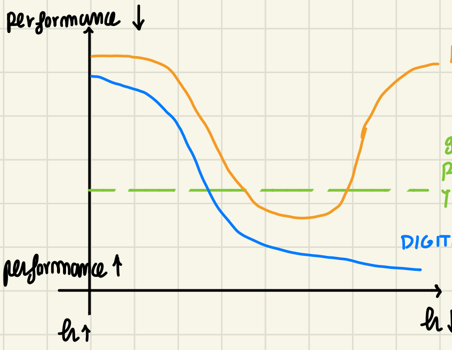
Finally the NETWORK performances can be caractenzed by:

- **EFFICENCY** $P_{EFF} = \frac{T_{TX}}{T_{DELAY}} \in (0,1)$
  - $P_{EFF} \to 0$ too high delay, BAD NET architecture
  - $P_{EFF} \to 1$ very good Network

- **UTILIZATION** $P_{UTIL} = \frac{T_{NET\ TRASMISSION}}{T_{TOTAL\ NET\ UTILIZ.}} \in (0,1)$
  - $P_{UTIL} \to 0$ NOT using resources
  - $P_{UTIL} \to 1$ risk of congestion

- **NETWORK STABILITY** related to the amount of unsent message in the node queue ≠ NCS stability

CRITICAL CHOICE, SAMPLING TIME h



for h decreasing, performances improves up to a certain h ... in fact for h too small, the requested packet to send per time unit increase too much, causing congestion and so destabilizing the NCS

when h decreases, performance tends to the continuous time one, we have a "finer" control action, more reactive and so more performing

CONTROL NETWORK { phisical layer + MAC PROTOCOL

- PROFIBUS (TDM/TP)
- control Net (TDM/TP)
- Device Net (CSMA/AMP)
- Ethernet (wireless) (CSMA/CD)

- DETERMINISTIC, based on scheduling and time division multiplexing (TDM)
  - Token bus / Token Ring (TP)
  - Master / slave
  - Time Division Multiple Access
- STOCHASTIC, based on carrier sense Multiple Access (CSMA) protocols
  - collision arbitration (CAN)
  - collision detection (Ethernet)

TDM PROTOCOLS

based on a deterministic allocation of available network resources, with a predetermined "sequence" of communication "turn" for each node

- TOKEN PASSING : each node talk when it has the token
  - pass the token to next one (on logical/phisical ring) when info transmitted or max time expires
- MASTER / SLAVE
- TIME DIVISION MULTIPLE ACCESS: Δt allocated $\forall$ node

(PRO) - communication guarantee
- No risk of monopolization by nodes
- delay bounded deterministically
- efficent integration of new nodes

(CONS) - time allocated for pass token can be relevant for net with lots of nodes
- low traffic loads

↳ that's why to have higher communication rate we can use CSMA protocols ⟹ (with huge collision risks..)

CSMA PROTOCOLS

based on a random access on the network, each node try to commmunicate on the BUS when he have information to send, if medium is a cable he listen to it if its free, if wireless somehow we arbitrate collision

- **CSMA / collision ARbitration** ⇒ CAN = CSMA / $\underbrace{\text{AMP}}$

Arbitration on Message Priority

Uses additional overhead (11bit) as ARBITRATION FIELD, to guarantee that packet with higher priority is sent!

(PRO) - optimal for short messages
- Transmission delay bound on prioritized message guarantee
- Priority message gain acess

(CONS) - slow data rate R
- Not good for big messages

- **CSMA / CD (collision detection)** ⇒ Ethernet { Hub based / Wireless

based on a random interframe + back off time, which decreases the possibility of collision when transmitting

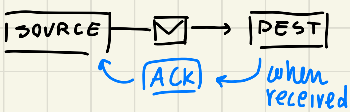
- **hub based**
  - If 10 collisions occurs TWAIT fixed
  - if 16 collisions occurs packet dropouts

(PRO) - no bandwith allocated for access to Net
- very good if low congestion

(CONS) - no guarantee of packet trasmission
- unbounded delay
- No prioritization
- big overhead, even for small Data size

- **Wireless**

based on trasmission of ACKnowledgement packets to ensure that packet reach destination

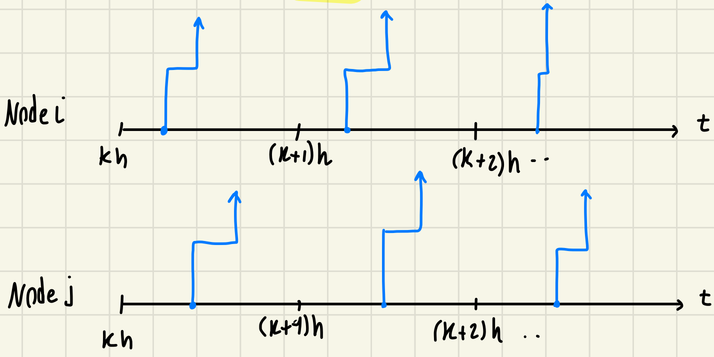


- lower actual throughput than the one "possible" because of big overhead and ACK

TIMING DIAGRAM

in general time varying delay, but for sure bounded

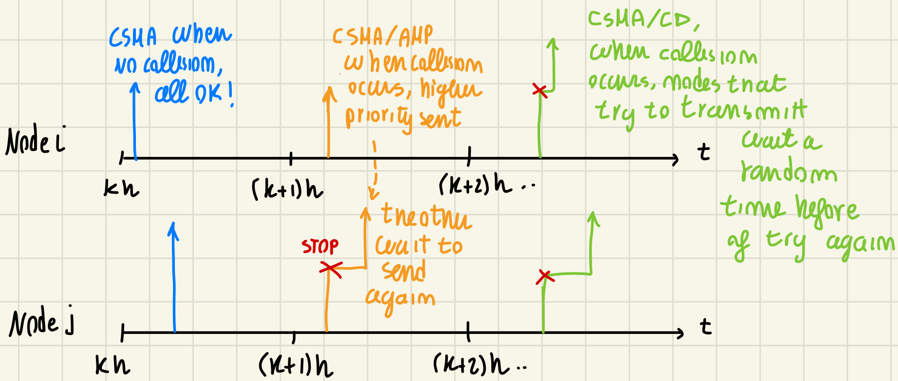
TDM



TDM/TP

when ready to sent wait for the token, than sent, never collision occurs because well partitioned BUS

CSMA

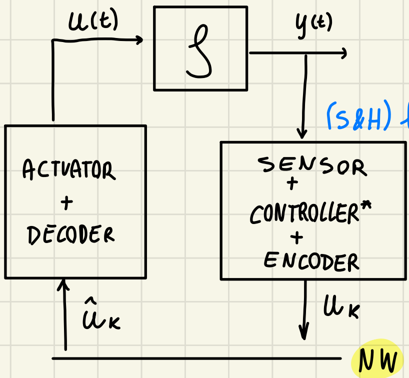


CONTROL UNDER DATA RATE LIMITATIONS

set-up: IDEAL NCS (No delay, No dropouts, No noise) ⇒ focus on the quantization issues, due to limited control action freedom (N_B), when we have the power to control system?

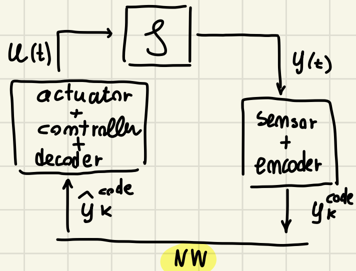
- extreme cases when $N_B = 1, 2, ..$ very limited, considering only data bits (NOT overhead ecc.)

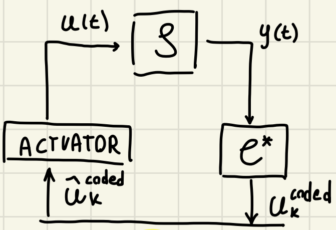
CONFIGURATION



⇒ but, all our reasoning can be done for the following schemes (with some adjustment)

- $\mathcal{C}$ co-located with ACTUATOR

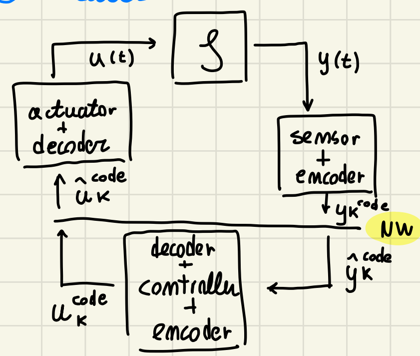




REFERENCE configuration $\mathcal{C}^*$ co-located with sensors

NW ~ ideal NW, $\hat{u}_K^{coded} = u_K^{coded}$

- $\mathcal{C}$ isolated



≠

DIGITAL CONTROL STRATEGY

- $u_K \in \mathbb{R}e$, unbounded, fine action
- $u_K = K x_K$ function control action
- NOTION of CONTROLLABILITY
  $\forall x_0 \; \exists u_0, u_1, \dots$ s.t $x_{\bar{k}} = 0$
- STABILIZABILITY
  $\forall x_0 \; \exists u_0, u_1, \dots$ s.t $x_K \xrightarrow[K \to +\infty]{} 0$
  (at least for unstable modes)

≠

IN NCS, under data rate limit

→ • $u_K \in \mathcal{U}$, $\mathcal{U}$ s.t $|\mathcal{U}| = 2^{N_B} = N$

→ • we should define notion of "selection function" to control in feedback

→ • No more a "global" stability notion and also No more convergence to a constant $\bar{x}$ equilibrium
(POSITIVELY INVARIANT SET and BOUNDABILITY)

Communication channel set-up

Let:
- R = data rate [bit/s], Given
- h = sampling time [s], user chosen
- N_B = number of bit per packet, user chosen
- ideal network

} given R we have to chose h, N_B to properly control the NCS

⇓

a first obvious constraint is:

[obviously, time required to "build" a packet should be smaller than the one required for sampling that information] ⇐ $(T_{FRAME} < h) \rightarrow \frac{N_B}{R} < h$

$$\frac{N_B}{h} < R$$

⟹ [trade-off for the choice of N_B, h]

IDEALLY { N_B big (finer discretization)
h small (reactive control action) } ⟹ but this calls for higher values of R, which is limited by Network

but that's NOT the only CONSTRAINT
to CONTROL the NCS, we wanna understand when the NCS can be properly controlled to guarantee some notion of "stability" (to be redefined for this systems)

QUESTIONS {
1) what new notion of controllability/stabilizability should be introduced for NCS (BOUNDABILITY)
2) for which condition this notion is guaranteed (Theorem on boundability)
3) IF system can be controlled as desired, how we control it? (selection function)
}

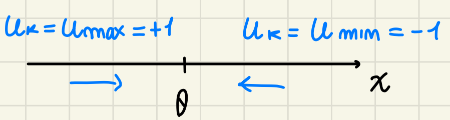
this arises from some example on a simple scalar system

$\mathcal{S}: \quad \dot{x}(t) = a x(t) + u(t) \longrightarrow$ controlled by $\mathcal{C}$ discrete, let's discretize $\rightarrow \begin{bmatrix} f = e^{ah} \\ g = \frac{b}{a}(e^{ah} - 1) \end{bmatrix} \quad x_{k+1} = f x_k + g u_k$

in our case u_k limited by N_B, to take $N = 2^{N_B}$ values

IF $N_B = 1 \qquad u_k \in \{u_{min}, u_{max}\} = \mathcal{U}$

a wise choice is to assign u_k values accordingly to current x_k (feedback) with a "smart" partition of X space



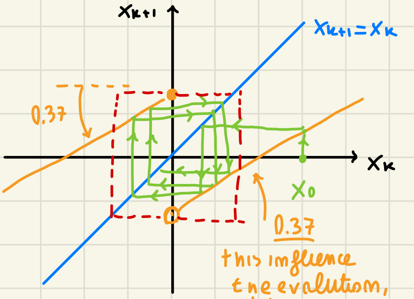
$$\mathcal{S}. \quad x_{k+1} = \begin{cases} f x_k + g u_{max} & x_k \leq 0 \\ f x_k + g u_{min} & x_k > 0 \end{cases}$$

this already hint that, $u_k \neq 0$ always, in no way $\bar{x}_k = 0$ converge (always acting on it) at this point, from a GRAPHICAL ANALYSIS, we can check on what happens for different "STABILITY" condition of the original $\mathcal{S}$ system, when our NCS is able to behave in a satisfactory way, and IF it "destabilize" our system

⇒ notice that we are dealing with a SWITCHING SYSTEM, stability behaviour has to be checked by GRAPHICAL ANALYSIS

$h = 1$ sec, $N_B = 1$ bit

1) $a = -1$ ($\mathcal{S}$ asympt stable) $\Rightarrow$ $x_{k+1} = h(x_k) = \begin{cases} 0.37 x_k + 0.63 & x_k \le 0 \\ 0.37 x_k - 0.63 & x_k > 0 \end{cases}$

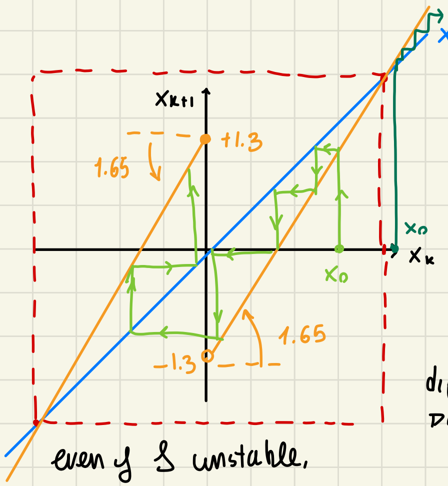


$\forall x_0,\ x_k \longrightarrow [-0.63, 0.63]$ PI SET

(h(xk) never encounter the blue line)

control action strong enough to maintain good bounded behaviour in a set

2) $a = +0.5$ ($\mathcal{S}$ small Instability) $\Rightarrow$ $x_{k+1} = h(x_k) = \begin{cases} 1.65 x_k + 1.3 & x_k \le 0 \\ 1.65 x_k - 1.3 & x_k > 0 \end{cases}$



encounter in

$x_k = 1.65 x_k + 1.3 \Rightarrow x_k = 2$

$\forall |x_0| > 2 \Rightarrow x_k \to +\infty$ diverge

$\forall |x_0| \le 2 \Rightarrow x_k \to [-2, 2]$

(LOCAL STABILITY)

↑

differences respect DIGITAL CONTROL →

⇓

$x_k \to [-1.3, 1.3]$ minimal PI set

converging to a set, NOT to a point

even if $\mathcal{S}$ unstable,

instability NOT strong enough, our freedom on the h, N_B choice allow control

3) $a = +1$ (high unstable) $\Rightarrow$ in this case graphically we can check that $\forall x_0,\ x_k \longrightarrow +\infty$

⇓

so, there is a new behaviour of state evolution (boundability) and somehow it is related on "a" level of instability respect N_B, h controller freedom

Let's answer our previous questions:

1) (new "stabilizability" notion in the control under data rate limit framework)

BOUNDABILITY

given $\mathcal{S}: \dot{x}(t) = Ax(t) + Bu_k$ with $u_k \in \mathcal{U} = \{\bar{u}_0, \bar{u}_1, \dots, \bar{u}_{N-1}\}$ $N = 2^{N_B}$

we say that the system is BOUNDABLE if

$\exists$ bounded set S and set M such that $M \subseteq S$ and such that:

IF $x_0 \in M$ than $\exists$ control sequence $u_0, u_1, \dots$ with $u_k \in \mathcal{U}$

s.t $x(t) \in S \ \forall t$

$\Downarrow$

2) when a system is boundable?

Theorem (SCALAR CASE)

$\exists \mathcal{U}$ s.t $\mathcal{S}: \dot{x} = ax + bu$ is BOUNDABLE

IFF

$$a \log_2 e \leq \frac{N_B}{h} < R$$

in fact in previous exampe

1) $a \log_2 e = -1.44 << 1$ bound.

2) $a \log_2 e = 0.72 < 1$ bound.

3) $a \log_2 e = 1.44 > 1$ unbound

(so, for sure if $a \log_2 e > R$, $\nexists \mathcal{U}$, no way to control our system boundability)

$\downarrow$

In general this theorem ensure that "$\exists \mathcal{U}$" but seem to NOT give us the responce to the last question, how to properly select u_k?

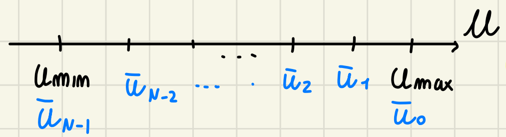
3) **SELECTION FUNCTION** Lukily, from proof of Theorem, a way to define the $\mathcal{U}$ set and selection function is given

$\Downarrow$

CASE a) at RATE LIMIT

when $\frac{N_B}{h} = a \log_2 e$ $\exists!$ selection function (only one..) given by a uniform partition of X, $\mathcal{U}$ $[N = 2^{N_B}]$

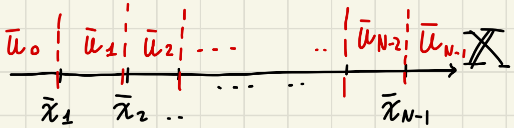
IF $u_k \in [u_{min}, u_{max}]$ known bound



$$\bar{u}_i = u_{max} - \frac{(u_{max} - u_{min})\, i}{N-1} \qquad i = 0, \dots, N-1$$

$\Downarrow$

selection function



$i = 0, \dots N$

$$\bar{x}_i = -\frac{b}{a} u_{max} + \frac{b}{a} (u_{max} - u_{min}) \frac{i}{N}$$

from this partition $\Rightarrow$

$$u_k = \begin{cases} \bar{u}_0 & \text{IF } x_k < \bar{x}_1 \\ \bar{u}_i & \text{IF } \bar{x}_i \le x_k < \bar{x}_{i+1} \\ \bar{u}_{N-1} & \text{IF } x_k > \bar{x}_{N-1} \end{cases}$$

Theorem ensure us that this selection function with this partition guarantee to have

$$\forall x_0 \in [\bar{x}_0, \bar{x}_N], \quad x_k \longrightarrow \underbrace{[\bar{x}_0, \bar{x}_N]}_{\text{this is a quite big set... NOT so performing}} \text{ PI set}$$

case b) ABOVE RATE LIMIT

when $\frac{N_B}{h} > a \log_2 e$

in this case we could use the same selection function of the rate limit case, but it can be improven, $[\bar{x}_0, \bar{x}_N]$ proves to be the maximal PI set (smaller one can be found)

$\Leftarrow$ maybe partition u, x space with NOT uniform sector, (logarithmic, ecc..)

- EXTENSION, VECTOR CASE

When we consider $\dot{x} = Ax + Bu \quad x \in \mathbb{R}^n \quad u \in \mathbb{R}$

[Necessary condition for BOUNDABILITY
IF $\mathcal{S}$ boundable $\Rightarrow$ (A, B) controllable]

Theorem still holds $\exists\, u$ s.t $\mathcal{S}$ is boundable IFF

$$a \log_2 e \le \frac{N_B}{h}$$

$$a = \sum_{i:\, \mathrm{Re}(\lambda_i(A)) \ge 0} \lambda_i$$

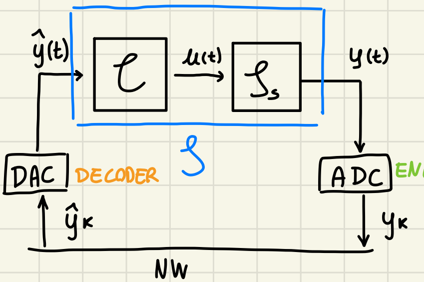
Theorem also provide a way to define the "bounding" selection function (complex)

Next step, remove the "IDEAL NETWORK" assumption, to check what happens to the NCS stability when we have delays / dropouts (new configuration) $\Longrightarrow$

EFFECT OF DELAY ON NCS

Now we focus on the consequence of delay only, assuming that N_B big enough to neglect quantization error, records = image $y_k = \hat{y}_k$

this analysis is done in a new reference configuration (general for many other NCS structures) ⇐



$\mathcal{C}$ co-located with actuator, $\mathcal{S}$ describe the $\hat{y}$ to y dynamic, considering that $\hat{y}$ arrive delayed respect y, with some consequence on overall NCS dynamic

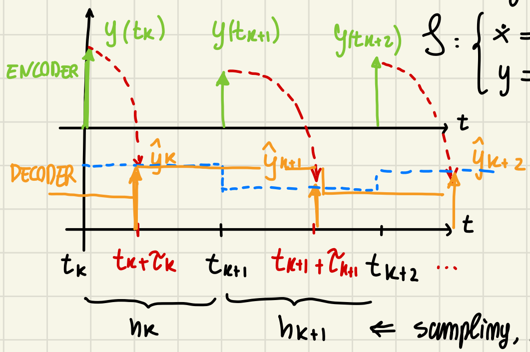
$$\mathcal{S}_s : \begin{cases} \dot{x}_s = A_s x_s + B_s u(t) \\ y(t) = C_s x_s \end{cases}$$

then defining properly $\mathcal{C}$ according to controller used (static output feedback, observer + state feedback, dynamic state feedback..) ⇒

we can bring to reference dyn:

$$\mathcal{S} : \begin{cases} \dot{x} = Ax + B\hat{y} \\ y = Cx \end{cases}$$

TIMING DIAGRAM



⇐ sampling, in general t-varying h_k

(S & H) → 0-ORDER HOLD

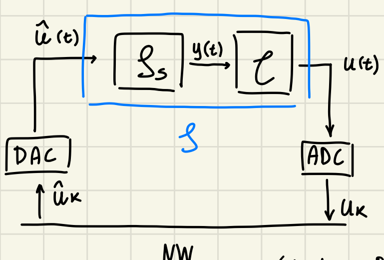
ideally $\hat{y}(t) = \hat{y}_k \quad t \in [t_k, t_{k+1}]$

reality $\hat{y}(t) = \begin{cases} \hat{y}_{k-1} & t \in [t_k, t_k + \tau_k] \\ \hat{y}_k & t \in [t_k + \tau_k, t_{k+1}] \end{cases}$

due to this unconsistent signal hold, in closed loop the system behaviour change inside h_k, affecting the dynamic

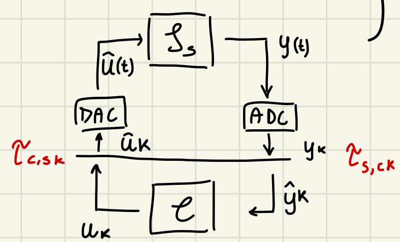
NOTICE, that $\mathcal{S}$ description can be adapted also for the following cases, just with a proper redefinition of A, B, C

- $\mathcal{C}$ co-located with sensor



$$\mathcal{S} : \begin{cases} \dot{x} = Ax(t) + B\hat{u}(t) \\ u(t) = Cx(t) \end{cases}$$

- $\mathcal{C}$ remote

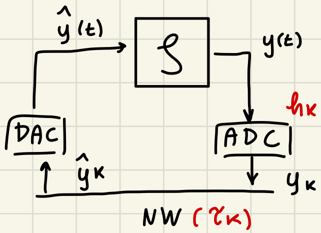


considering $\tau = \tau_{s,ck} + \tau_{c,sk}$

we can bring it to previous case

$$\mathcal{S} : \begin{cases} \dot{x} = Ax + B\hat{y} \\ y = Cx \end{cases}$$

SET-UP



$$S : \begin{cases} \dot{x} = Ax + B\hat{y} \\ y = Cx \end{cases}$$

and from now on we consider $\tau_K < h_K$ (Delay)

IF it occurs $\tau_K > h_K \Rightarrow$ DROPOUT, packet lost

$h_K = t_{K+1} - t_K$ time varying
τ_K time varying

$\rightarrow h, \tau$ (CASE 1)
$\rightarrow h_K, \tau_K$ (CASE 2)

we will see what change if
we analyze the stability of NCS when h, τ are
fixed or time-varying

$\Downarrow$

in general, first of all let's find out the dyn model of NCS

- when $\tau = 0$, $h_K = h$, without delay, from timing diagram $\hat{y}(t) = y_K \quad t \in [t_K, t_{K+1}]$

from LAGRANGE EQUATION

$$x_{K+1} = e^{A(t_{K+1}-t_K)} x_K + \int_{t_K}^{t_{K+1}} e^{A(t_{K+1}-\nu)} (B\hat{y}(\nu)) d\nu =$$

$A(t_{K+1}-t_K) = Ah$; $B y_K = BCx_K$ constant over ν

$$= \underbrace{\boxed{e^{Ah}}}_{F} x_K + \underbrace{\boxed{\Gamma(h)B}}_{G} C x_K = (F + GC) x_K$$

stability analyzed just through eig$(F+GC)$

- When $\tau_K \neq 0$, h_K time-varying

instead we should consider the fact that $\hat{y}(t) = \begin{cases} \hat{y}_{K-1} & t \in [t_K, t_K+\tau_K] \\ \hat{y}_K & t \in [t_K+\tau_K, t_{K+1}) \end{cases}$

$$x_{K+1} = e^{Ah_K} x_K + \underbrace{\left[\int_{t_K}^{t_K+\tau_K} e^{A(t_{K+1}-\nu)} d\nu \right]}_{e^{A(h_K-\tau_K)}\,\Gamma(\tau_K)} B\hat{y}_{K-1} + \underbrace{\left[\int_{t_K+\tau_K}^{t_{K+1}} e^{A(t_{K+1}-\nu)} d\nu \right]}_{\Gamma(h_K-\tau_K)} BC x_K$$

$\downarrow$

$$x_{K+1} = \left[e^{Ah_K} + \Gamma(h_K-\tau_K)BC \right] x_K + e^{A(h_K-\tau_K)} \Gamma(\tau_K) B \hat{y}_{K-1}$$

to bring it in state space form, we define a new state variable $z_K = \begin{bmatrix} x_K \\ \hat{y}_{K-1} \end{bmatrix}$

$$z_{K+1} = \begin{bmatrix} x_{K+1} \\ \hat{y}_K \end{bmatrix} = \underbrace{\begin{bmatrix} e^{Ah_K} + \Gamma(h_K-\tau_K)BC & e^{A(h_K-\tau_K)}\Gamma(\tau_K)B \\ C & 0 \end{bmatrix}}_{z_{K+1} = \Phi(h_K, \tau_K) z_K} \begin{bmatrix} x_K \\ \hat{y}_{K-1} \end{bmatrix}$$

Linear time-varying system (Parameter varying) from the found dynamic $z_{k+1} = \Phi(h_k, \tau_k)\, z_k$ (LTV / LPV system)

⇓

stability analysis becomes more complex whenever h_k, τ_k time-varying, we shoul relay on some new results, by LMI application..

CASE 1) when $\tau_k = \tau,\ h_k = h$ LTV ⇒ LTI

usual shur stability test on $\phi(h,\tau)$ are enough to ensure asympt stability of NCS dynamic ⇒ $z_{k+1} = \Phi(h,\tau)\, z_k$

$\Phi(h,\tau)$ shur stable ⟺ $\exists P > 0,\ P = P^T$ s.t $\Phi^T(h,\tau)\, P\, \Phi(h,\tau) - P < 0$
(NCS stable)

⇓

the stability analysis can be done also through:

- computation of $\Phi(h,\tau)$
- compute $P_\phi(\lambda) = \det(\lambda I - \Phi(h,\tau)) = \lambda^2 + a\lambda + b$
- Than relay on JURY CRITERION for shur stability, which in a 2ND ORDER polynomial becomes:

$$\Phi \text{ shur stable} \iff \begin{cases} -1 < b < 1 \\ -(1+b) < a < (1+b) \end{cases}$$

⇓

by some examples, we can see that accordingly to the value of h, τ the stability condition changes for a simple choice of a static state feedback control gain K

AND it could happen (NOT a rule) that, when

$\tau < h/2$ small delay → K even less conservative than in digital

$\tau > h/2$ big delay → NCS more conservative on K choice respect digital

even if it seems counter intuitive, it is a possibility..

CASE 2) when τ_k, h_k LTV system dynamic of NCS

in this case the analysis of the eigenvalues of $\Phi(h_k, \tau_k)\ \forall h_k, \tau_k$ is NOT enough! because it is a switching system, the dynamic depend on how system switch dynamic mode, it could even happen that switching between unstable modes, the overall dynamic is stable

NOTE (usefull later), when switching between two modes F_1, F_2 it could happen that

$$\begin{cases} x_{k+1} = F_1 x_k \\ x_{k+2} = F_2 x_{k+1} \end{cases} \rightsquigarrow x_{k+2} = \boxed{F_2 F_1} x_k$$

the dynamical evolution is different, in general $F_2F_1 \neq F_1F_2$

but $eig(F_1F_2) = eig(F_2F_1)$

$$\begin{cases} x_{k+1} = F_2 x_k \\ x_{k+2} = F_1 x_{k+1} \end{cases} \rightsquigarrow x_{k+2} = \boxed{F_1 F_2} x_k$$

but this observation is usefull when we JUST switch between two modes with a specific "pattern"

$\Downarrow$

in our case $\Phi(h_k, \tau_k)$ denote a switching behavior, the eigenvalue check is NOT enough but we can relay on a usefull result:

Theorem

the NCS described by $\Phi(h_k, \tau_k)$, assuming $h_k \in [h_{min}, h_{max}]$, $\tau_k \in [\tau_{min}, \tau_{max}]$ is asympt stable IF

$$\exists P = P^T \text{ s.t } \Phi(h_k, \tau_k)^T P \Phi(h, \tau) - P \prec 0$$

$$\left\| \begin{matrix} \forall h_k \in [h_{min}, h_{max}] \\ \forall \tau_k \in [\tau_{min}, \tau_{max}] \end{matrix} \right\|$$

only sufficent, because it restrict in finding one P matrix $\forall h_k, \tau_k$, so the condition is NOT the more representative... IF $\nexists P$ NO guarantee that the NCS is unstable...

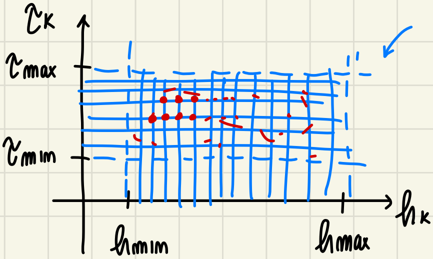
in principle this Theorem requires to prove inequality in an infinite amount of h_k, τ_k, NOT feasible in practice..

$\Downarrow$

this holds because if we define $V(z_k) = z_k^T P z_k$ Lyap theory proves to be true

to check Lyap inequality we relay on

GRIDDING METHOD



GRID the (h, τ) space and check for a finite h_k, τ_k, than relay on "continuity" to assume that in the inside points (h, τ) NOT checket, condition holds

$\Downarrow$

IF we check with this Theorem, we provide a "conservative" H, T range of h, τ where even if $h_k \in H$, $\tau_k \in T$ the NCS is asympt. stable, an usually it is smaller than the simple check on $\Phi(h, \tau)$ $\forall h, \tau$, here we have to find a common P matrix (Notice that we are NOT taking into account the rate of change of h, τ... it can be more/less conservative)

Note, for exercises When we analyze the stability of a NCS affected by delay, when we assume $h_k = h$ constant $\tau_k = \tau$ given and the control system fully described (K given)

⇓

we can be interested in computing the "MATI" = "MAXIMAL ALLOWED TRANSFER INTERVAL" (delay)

⇓

the h_{max} which guarantee stability for the given τ, K

↓

to do this, JUST leave $f = e^{ah}$ as the unknown, solve stability condition of NCS respect f and than find that h_{max}

EFFECT OF DROPOUTS ON NCS

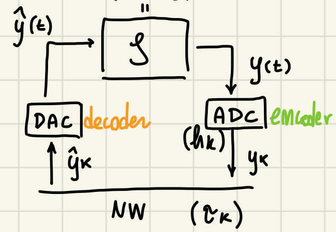
this time we deal with the stability effect of a packet dropout, which occurs when $\tau > h$, the datum become no more relayable, we discord it (OR if it is lost and never arrive at destination for some fault)

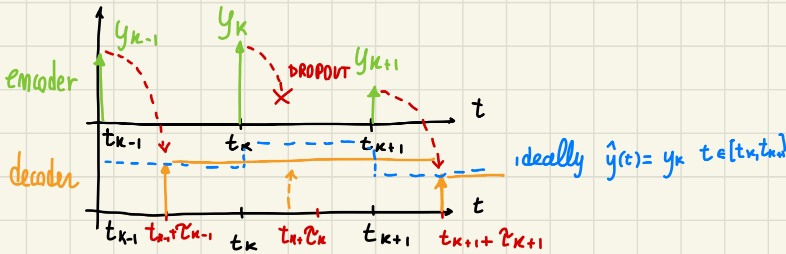
the approach is very similar to the one used in case of delay, we should define the NCS dynamic in this case...

REFERENCE CONFIGURATION

same as before, $\mathcal{C}$ co-located with actuator

$(\mathcal{C} + \mathcal{S}_s)$





here due to dropout $\hat{y}_k = \hat{y}_{k-1}$, kept until next datum arrival

Let's define $\vartheta_k = \{0, 1\}$ representative of dropout occurrence

$$\hat{y}_k = \vartheta_k y_k + (1-\vartheta_k)\hat{y}_{k-1} \quad (*)$$

$$\vartheta_k = \begin{cases} 1 & \text{if datum @ } t_k \text{ arrive} \\ 0 & \text{if datum @ } t_k \text{ get lost} \end{cases}$$

$\Downarrow$

by this definition, applyed on the dyn model computation (from previous result...) (use $\hat{y}_k$ model) $(*)$

$$x_{k+1} = e^{Ah_k} x_k + e^{A(h_k - \tau_k)}\Gamma(\tau_k) B \hat{y}_{k-1} + \Gamma(h_k - \tau_k) B \hat{y}_k$$

$$= \left[e^{Ah_k} + \vartheta_k \Gamma(h_k - \tau_k) BC\right] x_k + \left[e^{A(h_k-\tau_k)}\Gamma(\tau_k) + (1-\vartheta_k)\Gamma(h_k - \tau_k)\right] B \hat{y}_{k-1}$$

again, we can define the dyn in state space as

$$z_{k+1} = \begin{bmatrix} x_{k+1} \\ \hat{y}_k \end{bmatrix} = \underbrace{\begin{bmatrix} e^{Ah_k} + \vartheta_k \Gamma(h_k - \tau_k) BC & \left[e^{A(h_k-\tau_k)}\Gamma(\tau_k) + (1-\vartheta_k)\Gamma(h_k-\tau_k)\right] B \\ C\vartheta_k & (1-\vartheta_k) I \end{bmatrix}}_{} \begin{bmatrix} x_k \\ \hat{y}_{k-1} \end{bmatrix}$$

$$z_{k+1} = \tilde{\Phi}_{\vartheta_k}(h_k, \tau_k) z_k$$

in principle Φ now is function of δ_k, h_k, τ_k, we should relay on a 3D stability analysis...

↓ to make the analysis more affordable

HYPOTESIS $\tau_k = \tau,\ h_k = h$

$$z_{k+1} = \tilde{\Phi}_{\delta_k}(h,\tau) z_k$$

$\Rightarrow$ switching system between 2 modes

$$\begin{tikzcd}
z_{k+1} = \tilde{\Phi}_1(h,\tau) z_k \arrow[loop above, "\delta_{k+1}=1"] \arrow[r, bend left, "\delta_{k+1}=0"] & z_{k+1} = \tilde{\Phi}_0(h,\tau) z_k \arrow[loop above, "\delta_{k+1}=1"] \arrow[l, bend left, "\delta_{k+1}=1"]
\end{tikzcd}$$

(mode $\delta_k = 1$)

$\tilde{\Phi}_1(h,\tau) = \Phi(h,\tau)$ same as delay analysis matrix, here we can assume controlled properly, so $\tilde{\Phi}_1$ shall stable

(mode $\delta_k = 0$)

$$\tilde{\Phi}_0(h,\tau) = \begin{bmatrix} e^{Ah} & \left[e^{A(h-\tau)}\Gamma(\tau) + \Gamma(h-\tau)\right]B \\ 0 & I \end{bmatrix} = \begin{bmatrix} e^{Ah} & G \\ 0 & I \end{bmatrix} \to eig(\tilde{\Phi}_0) = eig(F) \cup 1$$

Never asympt. stable, at best its simply stable (it ereditate modes of open loop)

↓

Instead of using the classical Lyapunov Theory analysis, (NOT feasible because $\nexists P: \tilde{\Phi}_0^T P \tilde{\Phi}_0 - P < 0$) we relay on a characterization of the dropout (deterministic / stochastic) and on the switching mode behaviour

DETERMINISTIC FRAMEWORK

describe the dropout occurrency by a deterministic parameter "r" = dropout rate

$$\lim_{T\to\infty}\left(\frac{1}{T}\sum_{k=k_0}^{k_0+T-1}(1-\delta_k)\right) = r$$

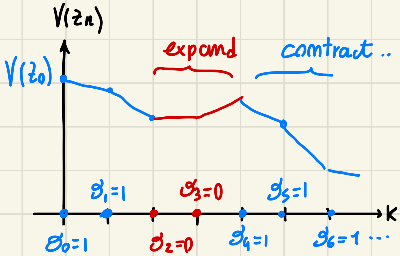
$$\begin{cases} r = 0 \text{ always trasmission} \to \text{all transmitt} \Rightarrow \tilde{\Phi}_1 \quad \text{GOOD} \\ r = 1 \text{ always dropout} \to \text{All lost} \Rightarrow \tilde{\Phi}_0 \quad \text{BAD} \end{cases}$$

↓

based on r defintion we can provide an idea on a Theorem that give us a way to check for asympt. stability of NCS in this framework we can find $V(z_k) = z_k^T P z_k$ at least good for $\tilde{\Phi}_1$ (good in sense of Lyap theory)

$\Downarrow$ such that

$$\begin{cases} \bullet \text{ for } \delta_k = 1 \; (\beta_1 \in (0,1)) & V(z_{k+1}) \le \underbrace{\beta_1}_{\text{contraction rate}} V(z_k) \\ \bullet \text{ for } \delta_k = 0 \; (\beta_0 > 1) & V(z_{k+1}) \le \underbrace{\beta_0}_{\text{expansion rate}} V(z_k) \end{cases}$$



$\Downarrow$

$$V(z_k) \le \beta_{\delta_{k-1}} V(z_{k-1}) \le \dots \le \beta_{\delta_{k-1}} \beta_{\delta_{k-2}} \cdots \beta_{\delta_0} V(z_0)$$

intuitively if it contract more than expand we can hope $V(z_k) \to 0$, so for Lyapunov we have stability

in an interval k, with a system characterized by $r \Rightarrow \begin{cases} \delta_1 \text{ occurs } (1-r)k \text{ times} \\ \delta_0 \text{ occurs } r \cdot k \text{ times} \end{cases}$

$$V(z_k) \le \beta_0^{rk} \beta_1^{(1-r)k} V(z_0) = (\beta_0^r \beta_1^{(1-r)})^k V(z_0) = \bar{\beta}^k V(z_0) \xrightarrow[\lim_{k \to +\infty}]{} \begin{cases} 0 & \text{if } \bar{\beta} < 1 \\ \infty & \text{if } \bar{\beta} > 1 \end{cases}$$

$\Downarrow$

Theorem

the NCS is asymptotically stable IF $\exists P = P^T > 0$ and $\beta_0 > 1$, $\beta_1 \in (0,1)$ s.t

$$\begin{cases} \tilde{\Phi}_0^T P \tilde{\Phi}_0 - P \le \beta_0 P \\ \tilde{\Phi}_1^T P \tilde{\Phi}_1 - P \le \beta_1 P \\ \beta_0^r \beta_1^{(1-r)} < 1 \end{cases}$$

← then we have to find P solving two LMI → numerically efficent

$\Uparrow$

← first we look for β_0, β_1 which respect this condition, by GRIDDING METHOD → numerically efficent

a usefull parameter we could define is $\bar{r}$, maximal dropout rate which is the r for which we lose stability

$$\beta_0^{\bar{r}} \beta_1^{(1-\bar{r})} = 1 \Rightarrow \bar{r} = -\frac{\ln(\beta_1)}{\ln(\beta_0) - \ln(\beta_1)}$$

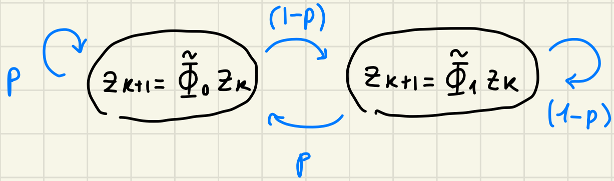
obviously $\bar{r} < 1$

this parameter could be used to understand how many packet we can decide to drop in case of congestion, without losing stability

STOCHASTIC FRAMEWORK

we model γ_k as a bernulli stochastic variable, so characterized by

$$p = \mathbb{P}(\gamma_k = 0) \qquad (1-p) = \mathbb{P}(\gamma_k = 1)$$



$\Rightarrow$ this is a MARKOV JUMP DECISION PROCESS (MJDP) model

(using Theory developed for this)

Theorem The NCS is mean square stable stable if $\exists\, Z = Z^T > 0$ s.t

$$\begin{bmatrix} Z & \sqrt{p}\,(\tilde{\Phi}_0 Z)^T & \sqrt{1-p}\,(\tilde{\Phi}_1 Z)^T \\ \sqrt{p}\,\tilde{\Phi}_0 Z & Z & 0 \\ \sqrt{1-p}\,\tilde{\Phi}_1 Z & 0 & Z \end{bmatrix} > 0$$

NCS mean square stable:

$$\begin{cases} \mathbb{E}[z_k] \xrightarrow{k \to +\infty} 0 & \text{(mean)} \\ \mathbb{E}[z_k z_k^T] \xrightarrow[k \to +\infty]{} 0 & \text{(correlation)} \end{cases}$$

$\downarrow$ also variance

strong result for stochastic processes

$\Uparrow$

notice that for

- $p = 0$ it is consistent with $\tilde{\Phi}_1^T P \tilde{\Phi}_1 - P < 0$
- $p = 1$ it is consistent with $\tilde{\Phi}_0^T P \tilde{\Phi}_0 - P < 0$ (never!)

also in this case it could be found $\bar{p}$ as maximum probability $(\gamma_k = 0)$ such that NCS is still mean square stable

NOTE, for exercises

when we have a given simple patterm $\gamma_k = 0, 1, 0, 1, 0 \dots$ that's a periodic dropout loss, system characterized by

$$\begin{cases} z_{2(k+1)} = \tilde{\Phi}_1 \tilde{\Phi}_0 z_k & (k \text{ even}) \\ z_{2(k+1)} = \tilde{\Phi}_0 \tilde{\Phi}_1 z_k & (k \text{ odd}) \end{cases}$$

but as stated previously $\tilde{\Phi}_1 \tilde{\Phi}_0 \neq \tilde{\Phi}_0 \tilde{\Phi}_1$, anyway

$$\operatorname{eig}(\tilde{\Phi}_1 \tilde{\Phi}_0) = \operatorname{eig}(\tilde{\Phi}_0 \tilde{\Phi}_1)$$

to analyze stability of NCS we just relay on shur stability of $\tilde{\Phi}_1 \tilde{\Phi}_0$ (OR $\tilde{\Phi}_0 \tilde{\Phi}_1$)